

Kinetic Video & Audio Production Pipeline

Autonomous 9:16 Remotion Composer & Edge-TTS Workflows

Executive Summary: This document provides actionable production implementations engineered by Pinnacle AI Solutions. Use this blueprint to architect zero-downtime, autonomous AI agent pipelines that streamline operational scale and client acquisition.

1. Automated 9:16 Short-Form Video Doctrine

Short-form video is the highest-converting organic channel for AI solution agencies. Pinnacle AI Solutions employs a programmatic Remotion composer combined with Edge-TTS and Kokoro neural audio engines to render viral 9:16 vertical videos in minutes without manual timeline editing.

The pipeline strips extraneous media, extracts core concepts, synthesizes broadcast-quality voiceovers, and overlays kinetic word-by-word captions in high-retention vertical safe zones.

- Format: Full-bleed 1080x1920 (9:16) at 30 FPS.
- Hook Velocity: Dynamic motion visual within the first 2.8 seconds.
- Voice Standard: en-US-AvaNeural (+12% rate) or Kokoro af_heart at -14 LUFS integrated output.

2. Sound Design & Retention Engineering

Sound design dictates retention. Background audio beds must be frequency-checked against the human vocal range (300Hz - 3400Hz) to prevent acoustic clash. Pinnacle AI mixes background beds at volume=0.32 and sound effects normalized to -16 LUFS.

- Kinetic Captions: Positioned at y:1680px (safe zone above platform UI) with electric gold highlight (#FACC15).
- Zero Generic AI Slop: Authentic UI screencasts, keynote demos, and vector typography only.
- Automated Desktop Delivery: All renders delivered directly to Windows Desktop folders.

```
// Remotion Audio Bed Mixing Equation
src={staticFile("audio/yt_library_ambient_pulse.mp3")}
volume={(f) => interpolate(f, [0, 30, duration - 30, duration], [0, 0.32, 0.32, 0])}
/>
```